Can a regression discontinuity identify the ICLR venue premium?

A design memo for CitesBench

2026-08-20

Why we need the premium at all

CitesBench scores a selection regime by the citations of the papers it admits. The problem is that acceptance itself raises citations. Write

$$Y_i = Y_i(0) + D_i\tau$$

where Y_i is realised log citations, D_i is historical ICLR acceptance, and τ is the venue premium.

The area-chair regime's admitted set is, by definition, $\{i : D_i = 1\}$. So it collects $Y_i(0) + \tau$ on every paper it picks. An LLM regime collects only $Y_i(0)$ on any paper the humans rejected. The comparison is therefore biased **toward the humans**, by roughly τ times the share of the LLM's slate that the humans had rejected.

This has a useful consequence and an awkward one.

- **Useful.** "The LLM matches or beats the humans" is *conservative* under this bias. We can claim it without knowing τ.
- **Awkward.** "The humans beat the LLM" is not identified at all without τ. Neither is any statement about the size of a gap.

So we want τ. A fuzzy regression discontinuity on the mean reviewer score is the obvious instinct, and it is the right instinct. This memo is about whether the data supports it.

What the design requires

A fuzzy RD needs four things. Fuzziness relaxes only the second one, from "treatment is determined by the threshold" to "treatment probability jumps at the threshold".

1. **A threshold in the assignment rule.** Some institutional cutoff that actually governs decisions.
2. **A jump in the probability of treatment at that threshold.**
3. **Data arbitrarily close to the threshold on both sides**, because the estimate is a limit taken from each side.
4. **An exclusion restriction.** Crossing the threshold affects citations *only* through acceptance.

Take these in turn against ICLR 2018–2020, with the running variable $z_i = \text{mean review score}_i - \text{cutoff}_y$.

What the data looks like

Support: there is almost nothing at the threshold

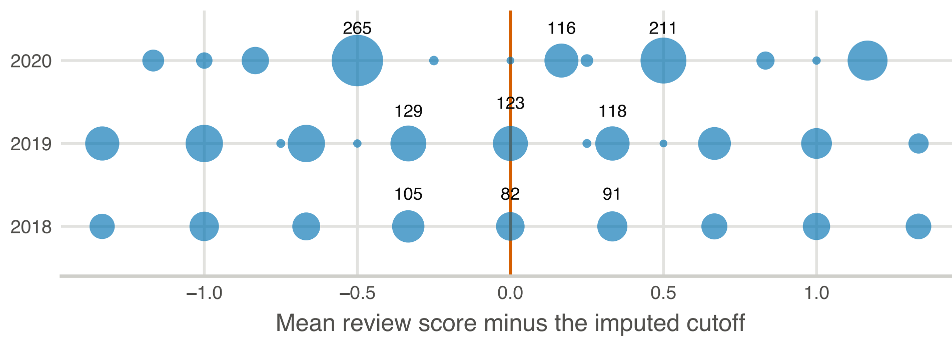

Figure 1: Mass points of the running variable. Bubble area is the number of papers; labels show counts near the cutoff. The vertical line is the imputed threshold.

The score is the mean of integer reviewer ratings, and 4,213 of 4,554 papers have exactly three reviews. So the running variable can only land on multiples of $\frac{1}{3}$. It takes 29 to 41 distinct values per year, and near the cutoff it collapses to a handful of clumps.

year	within 0.10 of cutoff	within 0.25	distinct values
2018	85	86	29
2019	123	123	41
2020	**6**	122	39

Two things to read off this. In 2018 and 2019, essentially every paper "near" the cutoff sits *exactly on it*, and the next mass is a third of a point away. In 2020 there are six papers within 0.1, with 265 and 211 papers parked at ±0.5.

Requirement 3 fails outright. There is no neighbourhood of the threshold to take a limit through. A local linear fit is not interpolating nearby data; it is extrapolating from clumps a third to a half of a rating point away, and the answer is whatever the slope assumption says it is. Kolesár and Rothe (2018) show that conventional RD inference is invalid in exactly this case.

First stage: a ramp with a small step in it

This is the panel that decides the question, and it is more interesting than a flat "no first stage".

There *is* a jump at zero. Fitting local linear with separate slopes gives a first-stage discontinuity of 0.18 to 0.26 depending on bandwidth, and it is not zero. An earlier version of this analysis claimed there was no first stage; that was wrong, and the data said so.

But look at the shape. $P(\text{accept})$ climbs smoothly from near 0% to near 100% across the whole range. In 2018 it runs 0%, 5%, 8%, 34%, 57%, 75%, 95%, 100%. The increment *across* the cutoff, 8% to 34%, is the same order as the increment one step above it, 34% to 57%. The threshold is not special. It is one step on a sigmoid.

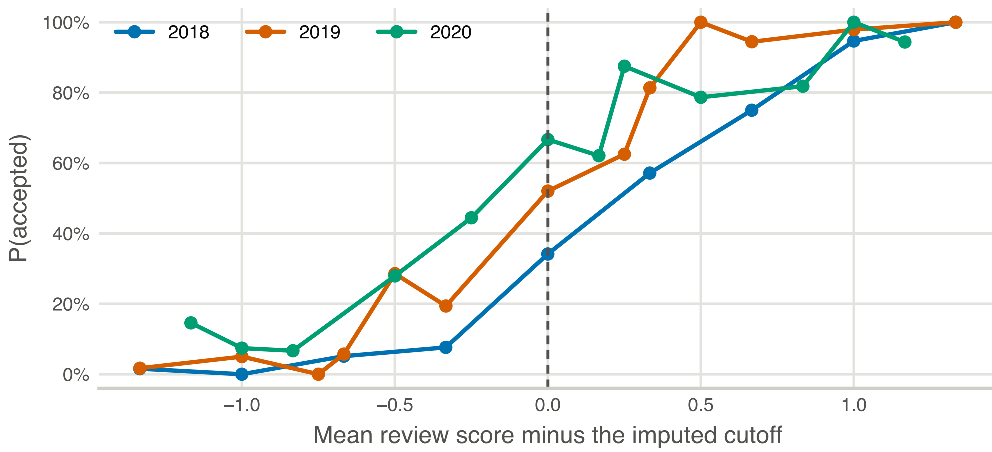

Figure 2: Probability of acceptance at each mass point of the running variable, by year. Points with fewer than five papers are suppressed.

That is what you should expect from the institution. Area chairs read a continuous quality signal and exercise discretion. Discretion applied to a continuous signal produces a smooth propensity. A genuine jump needs a *rule*, and ICLR publishes none.

Stability: the estimate is a choice of bandwidth

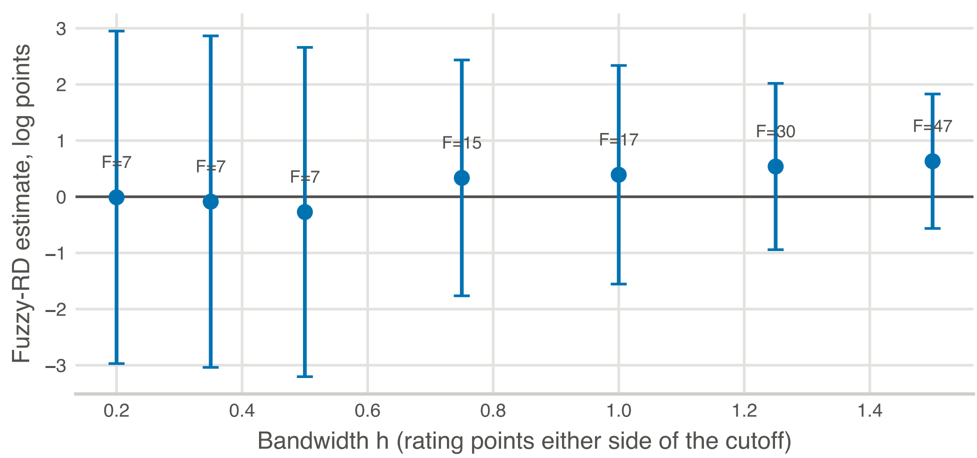

Figure 3: Fuzzy-RD Wald estimate against bandwidth, with 95% confidence intervals and the first-stage F at each bandwidth.

h	n	first-stage jump	F	estimate	95% CI
0.20	323	0.184	7.0	−0.01	[−2.97, 2.95]
0.35	787	0.182	7.1	−0.09	[−3.04, 2.86]
0.50	1,270	0.181	7.4	−0.27	[−3.20, 2.66]
0.75	1,669	0.219	14.6	+0.34	[−1.76, 2.44]

h	n	first-stage jump	F	estimate	95% CI
1.00	2,116	0.216	17.3	+0.39	[−1.55, 2.34]
1.25	2,417	0.236	30.2	+0.54	[−0.94, 2.02]
1.50	3,045	0.261	46.6	+0.63	[−0.56, 1.83]

Three readings, in order of how damaging they are.

Every interval contains zero, and every interval is enormous. The narrowest is $[-0.56, 1.83]$ log points. A premium of zero and a premium that multiplies citations sixfold are both inside it. Whatever else is true, this design does not deliver a number we could put in a sensitivity analysis, let alone an abstract.

The estimate does not converge as $h \to 0$. It marches monotonically from -0.27 to $+0.63$ as the window widens, a span of 0.9 log points. RD asymptotics run the other way: bias shrinks as $h \to 0$, so the small-h estimates should be the trustworthy ones. Those are the ones sitting at zero with ± 3 around them. Reporting any single number here would be reporting a bandwidth choice.

F **rises with bandwidth.** This is the tell, and it is the sentence to use in the meeting. A real discontinuity does not become *easier* to detect as you look further away from it. F climbing from 7 to 47 as h goes from 0.20 to 1.50 is the signature of a step function absorbing a slope: the wider the window, the more of the sigmoid's gradient the "above cutoff" dummy can soak up. It is measuring the slope, not a jump.

Two problems that more data would not fix

The above is about power and functional form. These two are structural.

The cutoff was reverse-engineered from the decisions. The value arrives pre-computed in `data/OpenAlex/openalex_rdd_arxiv_paper_level.csv`; no script in this repository derives it. It was fitted to the realised accept/reject pattern. So the "threshold" is a function of the outcome it is supposed to instrument, and the density test failing at the cutoff (+84.6%) partly reflects that it was placed where the mass already sat. Requirement 1 is not satisfied by an estimated cutoff.

The exclusion restriction fails independently. ICLR reviews and scores are public on OpenReview, for rejected papers as well as accepted ones. That is how this dataset exists at all. So crossing the threshold changes a *publicly visible quality signal*, not only the acceptance decision. A reader can see 6.0 rather than 5.67. Requirement 4 asks that the instrument work only through D, and here it plainly has a second channel.

Why the "working" estimate in the older output is not one

`outputs/fuzzy_rdd.md` reports a healthy-looking local *constant* specification: F of 172 to 674, estimate 0.94 to 1.01. It is worth knowing why that is not a rescue.

Dropping the slope does not remove the slope from the data. It reassigns the entire gradient across the bandwidth to a "jump" at the centre. With a smooth propensity, that estimate is a difference in means between papers a third of a rating point apart, which is a matched comparison and not a discontinuity. It inherits every confounder that varies over a third of a rating point, and reviewer scores are informative about quality, so plenty does.

The strong F is a symptom of the misspecification, not evidence against it.

What would make an RD work here

Being fair to the design, since it is the right instinct:

- A venue with a **published numeric threshold** that area chairs are instructed to follow. Some journals and grant panels have this. ICLR does not.
- A **continuous** running variable, or at least one with support near the cutoff. Many more reviews per paper would do it; three integers will not.
- A cutoff **fixed ex ante and documented**, not fitted afterwards.

None of the three is available for ICLR 2018–2020. This is a property of the institution and the data, not of the estimation.

What to do instead

Three routes. The first is the one I would put in the paper.

1. Treat τ as a sensitivity parameter, and report the breakdown value. Do not estimate the premium. Report the regime contrast Δ as a function of τ and mark the value at which each conclusion flips. The sentence becomes: *the LLM's advantage survives any venue premium below τ^*, and the literature puts plausible premia in range R.* This converts an unidentifiable nuisance into a stated robustness range, which is standard practice when a parameter is not recoverable. It also plays to the bias direction from the first section: the main claim is already conservative, so the breakdown value is likely to be comfortably large.

2. Matched comparison among the rejected. Rejected papers that later appeared elsewhere, against accepted ICLR papers, matched on score and year. This is selection-on-observables, so a weaker assumption than RD, but it is an assumption we can state and probe rather than one that is silently violated. The data supports it today.

3. Honest confidence intervals for a discrete running variable. Kolesár–Rothe if we want to keep the RD framing. It requires a curvature bound we would have to argue for, the intervals will be wide, and it does not address either structural problem above. Defensible as an appendix, not as the headline.

The short version

The RD instinct is correct, and the failure is not weak data. A fuzzy RD needs a threshold that governs decisions; ICLR area chairs apply discretion to a continuous score, so the acceptance probability is a smooth curve rather than a step. The small jump we do detect is bandwidth-dependent, the first-stage F rises with bandwidth in the way misspecification produces, the estimate never settles as the window shrinks, and every confidence interval spans zero and several log points. On top of that the cutoff was fitted to the decisions it would instrument, and public OpenReview scores give the instrument a second channel to the outcome.

We should report the venue premium as a sensitivity range with a breakdown value, not as an estimate.

Produced by `python src/analysis/rdd_diagnostic.py`, on the canonical citation table. The older `fuzzy_rdd.py` uses the superseded OpenAlex pull (26.3 pp accept/reject coverage differential, against 3.9 pp now).